Newton's Laws of Motion

Sir Isaac Newton formulated three fundamental laws of motion that describe the relationship between any physical object and the forces acting upon it.

First Law: An object at rest remains at rest, and an object in motion remains in motion at constant velocity unless acted upon by a net external force (Inertia).

Second Law: The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass, expressed mathematically as F = ma.
Force produces acceleration in the direction of the applied force.

Third Law: For every action, there is an equal and opposite reaction. Forces always occur in pairs.

Conservation of Energy and Momentum

The principle of conservation of energy states that energy cannot be created or destroyed, only transformed from one form to another. In an isolated system, the total mechanical energy remains constant: E = K + U (Kinetic plus Potential energy).

Linear momentum is defined as p = mv (mass multiplied by velocity). In the absence of an external net force, the total momentum of a closed system is conserved across collisions.

Impulse is the integral of force over time (J = F * delta_t), which equals the change in momentum (delta_p).

Gravitation and Planetary Motion

Newton's Universal Law of Gravitation describes the attractive force between any two masses:
$F_g = G * (m1 * m2) / r^2$, where G is the gravitational constant.

Johannes Kepler formulated empirical laws of planetary motion:
Kepler's first law states that the orbit of a planet is an ellipse with the Sun situated at one of the two foci.

Kepler's second law (Law of Equal Areas) states that a line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time.

Kepler's third law relates the orbital period T and semi-major axis a: T^2 is proportional to a^3.